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Assembly structure and lighting device

Abstract

A lighting device comprises a first component, a second component, at least one assembling module, at least one light guide plate, and at least one light-emitting unit. The assembling module is mounted at the second component and comprises an elastic element and at two engaging elements, which are moveable along an abutting direction. The elastic element pushes the engaging elements along the abutting direction, such that the engaging elements extend out from the second component and abut against at least one matching portion of the first component along the abutting direction to detachably assemble the second component to the first component along an assembling direction, which is nonparallel to the abutting direction. The at least one light guide plate is mounted at the first component, and the at least one light-emitting unit is mounted at one of the second component and the engaging element of the assembling module and faces the light guide plate.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an assembly structure and a lighting device having the assembly structure, especially to the assembly structure allowing a first component and a second component to be detachably assembled.

2. Description of Related Art

(2) A conventional flat lighting device has a frame, a light guide plate, multiple light-emitting units, a driver, etc. The frame is mounted at the ceiling. The light guide plate and the light-emitting units are mounted in the frame. The driver is mounted at a back side of the light guide plate for driving the light-emitting units to emit light.

(3) When the conventional flat lighting device stops working, it might be because either the light-emitting units or the driver is broken. However, maintenance staffs cannot determine which element causes the malfunction on-site, and the light-emitting units and the driver are hard to be detached from the conventional flat lighting device. Therefore, the on-site maintenance staff has to replace a whole set of the conventional flat lighting device with a new set, and then brings the malfunctioning flat lighting device back for further trouble-shootings to confirm which element causes the malfunction. To demount the whole set of the conventional flat lighting device is inconvenient and time-consuming, and the whole set of the flat lighting device is large in volume and thus hard to be carried. Furthermore, when the conventional flat lighting device malfunctions or is broken, since it is hard to be detached for partial replacement, the common practice is to throw the whole set of the conventional flat lighting device away and replace it with a new one, causing unnecessary wasting, and thus needs to be improved.

SUMMARY OF THE INVENTION

(4) The main objective of the present invention is to provide a lighting device to resolve the high time consumption and inconvenience that when malfunctioning, the whole set of the conventional flat lighting device needs to be detached since the light-emitting units and the driver are hard to be detached.

(5) The lighting device comprises a first component, a second component, at least one assembling module, at least one light guide plate, and at least one light-emitting unit. The first component has at least one matching portion. The at least one assembling module is mounted at the second component and facilitates the second component to be detachably assembled to the first component along an assembling direction. The at least one assembling module comprises two engaging elements and an elastic element, and the two engaging elements are disposed at two sides opposite to each other of the second component along the abutting direction. The at least one engaging element is movably mounted at the second component along an abutting direction. The elastic element is mounted in the second component and disposed between the two engaging elements, and is capable of pushing the two engaging elements along the abutting direction, such that one of the two engaging elements abuts against the at least one matching portion of the first component along the abutting direction to assemble the second component to the first component. The assembling direction and the abutting direction are nonparallel. The at least one light guide plate is mounted at the first component, and the at least one light-emitting unit is mounted at one of the second component and the two engaging elements, and is faced toward the at least one light guide plate.

(6) The at least one assembling module of the assembly structure allows the second component to be detachably assembled with the first component. When the second component extends into the first component along the assembling direction and is adjacent to the at least one matching portion of the first component, the elastic element abuts against the two engaging elements along the abutting direction, so one of the two engaging elements abuts against the at least one matching portion of the first component along the abutting direction to assemble the second component to the

first component. Detaching the two engaging elements along a direction opposite to the abutting direction from the at least one matching portion, the second component can be detached.

(7) Each of the two engaging elements is pushed outwardly along the abutting direction by the elastic element, and one of the two engaging elements abuts against the at least one matching portion of the first component, thereby engaging the first and the second components. To quickly detach the second component, by applying a force on each of the two engaging elements to compress the elastic element inwardly along the abutting direction, each of the two engaging elements will no longer abut against the at least one matching portion of the first component. Therefore, the first and the second components will be no longer engaged together, facilitating the next detaching process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an enlarged exploded perspective view of an embodiment of an assembly structure in accordance with the present invention;
- (2) FIG. 2 is another enlarged exploded perspective view of the assembly structure in FIG. 1;
- (3) FIG. 3 is an enlarged top view of the assembly structure in FIG. 1, before an actuator is operated;
- (4) FIG. 4 is a front sectional view of the assembly structure in FIG. 1, when an elastic element is pressed outwardly;
- (5) FIG. 5 is an enlarged perspective view of the assembly structure in FIG. 1, before the actuator is operated;
- (6) FIG. 6 is an enlarged perspective view of the assembly structure in FIG. 1, after the actuator is operated;
- (7) FIG. 7 is an enlarged top operating view of the assembly structure in FIG. 1, after the actuator is operated;
- (8) FIG. 8 is a front sectional view of the assembly structure in FIG. 1, when the elastic element is compressed inwardly;
- (9) FIG. 9 is a front view of a second component and at least one assembling module of the assembly structure in FIG. 1;
- (10) FIG. 10 is an exploded perspective view of the second component and the at least one assembling module of the assembly structure in FIG. 1;
- (11) FIG. 11 is a perspective view of an embodiment of a lighting device in accordance with the present invention; and
- (12) FIG. 12 depicts the lighting device in FIG. 11, wherein the second component, the at least one assembling module, at least one light-emitting unit and a driver are detached from a first component.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (13) With reference to FIGS. 1 and 2, a preferable embodiment of an assembly structure in accordance with the present invention comprises a first component 10, a second component 20 and at least one assembling module 30.
- (14) With reference to FIGS. 1 and 2, the first component 10 has at least one matching portion 11.
- (15) With reference to FIGS. 1 and 2, the at least one assembling module 30 is mounted at the second component 20 and facilitates the second component 20 to be detachably assembled to the first component 10 along an assembling direction D1. With reference to FIGS. 1, 3 and 4, the at least one assembling module 30 comprises at least one engaging element 31 and an elastic element 32. The at least one engaging element 31 is movably and detachably mounted at the second component 20 along an abutting direction D2. The elastic element 32 is mounted in the second

component **20** and pushes the at least one engaging element **31** along the abutting direction **D2**, such that the at least one engaging element **31** abuts against the at least one matching portion **11** of the first component **10** along the abutting direction **D2** to assemble the second component **20** to the first component **10**. The assembling direction **D1** and the abutting direction **D2** are nonparallel.

(16) With reference to FIGS. **1** to **3**, the at least one assembling module **30** of the assembly structure facilitates the second component **20** to be attached to or detached from the first component **10**. When the second component **20** extends into the first component **10** along the assembling direction **D1** and is adjacent to the at least one matching portion **11** of the first component **10**, the elastic element **32** pushes the at least one engaging element **31** along the abutting direction **D2**. So, the at least one engaging element **31** abuts against the at least one matching portion **11** of the first component **10** along the abutting direction **D2** to assemble the second component **20** to the first component **10**. Further, with reference to FIGS. **7** and **8**, to quickly detach the first and second components, by forcing the at least one engaging element **31** to compress the elastic element **32** inwardly along the abutting direction **D2**, the at least one engaging element **31** will be moved away and detached from the at least one matching portion **11**. Therefore, the first component **10** and the second component **20** will no longer be fixed and engaged together, and thus the second component **20** can be detached to facilitate the next detaching process. So, the second component **20** can be easily attached to and detached from the first component **10**.

(17) With reference to FIG. **2**, to compress the elastic element **32** by operating the engaging element **31** more easily, the at least one assembling module **30** of the embodiment further comprises an actuator **33**. The actuator **33** is mounted at the second component **20** and can be operated to move along a force-applying direction **D3**. As shown in FIGS. **5** to **7**, when the actuator **33** is operated to move along the force-applying direction **D3**, the actuator **33** will push the at least one engaging element **31** to depart from the at least one matching portion **11** along the abutting direction **D2**. So, the second component **20** can be detached from the first component **10** along a direction, which is opposite to the assembling direction **D1**. Operating the actuator **33** facilitates moving the at least one engaging element **31** away from the at least one matching portion **11**, therefore increasing the convenience of detaching the second component **20**. Besides, the assembling direction **D1**, the abutting direction **D2** and the force-applying direction **D3** are preferably perpendicular to each other to facilitate ease in manufacturing the assembly structure.

(18) With reference to FIGS. **3**, **7** and **8**, to smooth the at least one engaging element **31**'s movement while pushing, the actuator **33** of the embodiment has at least one first guiding inclined surface **331**, and the at least one engaging element **31** has a second guiding inclined surface **311**. The second guiding inclined surface **311** abuts against the at least one first guiding inclined surface **331**. When the actuator **33** is operated to move along the force-applying direction **D3**, the at least one first guiding inclined surface **331** of the actuator **33** pushes the second guiding inclined surface **311** of the at least one engaging element **31**. The at least one first guiding inclined surface **331** and the second guiding inclined surface **311** both incline relative to the force-applying direction **D3** and to the abutting direction **D2**. While the actuator **33** is operated, the at least one first guiding inclined surface **331** of the actuator **33** pushes the second guiding inclined surface **311** of the at least one engaging element **31**, thereby guiding the at least one engaging element **31** to smoothly move away from the at least one matching portion **11** along the abutting direction **D2** to compress the elastic element **32** inwardly. When the actuator **33** is not operated, the elastic element **32** restores from compression and pushes the at least one engaging element **31** to move toward the at least one matching portion **11**. The at least one engaging element **31** then pushes the actuator **33** back to its original position which is prior-to-operation, through the second guiding inclined surface **311**, so manually restoring the actuator **33** is unnecessary.

(19) Preferably, as shown in FIGS. **1** and **9**, the at least one assembling module **30** comprises a position-limiting plate **34**, which is mounted on and covers the second component **20**. The at least one engaging element **31** and the elastic element **32** are disposed at an inner side of the position-

limiting plate **34**. The at least one engaging element **31**, the elastic element **32** and the actuator **33** are limited in position along the assembling direction **D1** by the position-limiting plate **34** and the second component **20**. The position-limiting plate **34** has a position-limiting hole **341** elongated along the force-applying direction **D3**. The actuator **33** has a position-limiting hook **332**, which extends out from the position-limiting hole **341** and abuts against an outer side of the position-limiting plate **34**, and therefore the actuator **33** can only be moved along the force-applying direction **D3** to keep the operating stability. In addition, the position-limiting plate **34** further has a covering panel **342** elongated inwardly. The covering panel **342** limits the at least one engaging element **31** in position along the force-applying direction **D3**, thereby increasing operating and installing stabilities of the at least one engaging element **31**.

(20) Additionally, with reference to FIGS. **9** and **10**, the actuator **33** has a handle **333** and a guiding portion **334**. The handle **333** extends out from a bottom side of the second component **20**. The guiding portion **334** is disposed in the second component **20**. The at least one first guiding inclined surface **331** is formed at the guiding portion **334**. The handle **333** facilitates a user to operate the handle **333** from the bottom side of the second component **20**, hence enhancing convenience in use. Besides, the second component **20** has an opening hole **21** formed at the bottom side of the second component **20**. The handle **333** extends out from the opening hole **21** of the second component **20**. The second component **20** has at least one positioning protrusion **211** formed at a periphery of the opening hole **21**. The actuator **33** has at least one positioning recess **335** adjacent to the handle **333**. The at least one positioning protrusion **211** extends into the at least one positioning recess **335** to limit the actuator in position along the assembling direction **D1**. Therefore, when the actuator **33** is operated to move along the force-applying direction **D3**, the actuator **33** will not shift away along the assembling direction **D1**, so the connecting stability between the actuator **33** and the second component **20** is increased.

(21) Furthermore, with reference to FIGS. **5**, **6** and **10**, the actuator **33** has a fixing portion **336** extending along the force-applying direction **D3** toward the first component **10** and extending into the first component **10** along the force-applying direction **D3**. That is, the fixing portion **336** is elongated toward a side of the actuator **33** far away from the first guiding inclined surface **331**. When the second component **20** is connected to the first component **10**, the fixing portion **336** extends into the first component **10**, such that the second component **20** is limited in position by both of the at least one engaging element **31** and the fixing portion **336** of the actuator **33** simultaneously. When the actuator **33** is operated and moved along the force-applying direction **D3**, the fixing portion **336** will be detached from the first component **10**. By the fixing portion **336**, the connecting stability between the second component **20** and the first component **10** is increased. Furthermore, a top side of the fixing portion **336** forms an arched edge, which can guide the fixing portion **336**, so the fixing portion **336** extends into the first component **10** more easily, thereby helping the second component **20** to be positioned and connected to the first component **10**.

(22) As shown in FIG. **10**, the at least one engaging element **31** has an engaging chunk **312** and a guiding chunk **313**. The engaging chunk **312** is elongated along the force-applying direction **D3** and abuts against the at least one matching portion **11** of the first component **10**. The guiding chunk **313** is mounted at the engaging chunk **312**. The second guiding inclined surface **311** is formed at the guiding chunk **313**. The engaging chunk **312** and the guiding chunk **313** can be one-piece formed or be two separate units that are assembled together. Preferably, as shown in FIG. **8**, the engaging chunk **312** has a chunk guiding surface **314**, which is elongated along the force-applying direction **D3** and inclines with respect to the assembling direction **D1**. The chunk guiding surface **314** abuts against the at least one matching portion **11** of the first component **10**. While assembling the second component **20** to the first component **10**, the chunk guiding surface **314** of the engaging chunk **312** guides the second component **20** to a stable assembling position such that the at least one engaging element **31** actually abuts against the at least one matching portion **11** of the first component **10**, thereby increasing the connecting stability between the first component **10** and the

second component **20**.

(23) In an embodiment of the assembly structure in accordance with the present invention, with reference to FIG. 3, the first component **10** has two said matching portions **11**, and the at least one assembling module **30** comprises two said engaging elements **31**. The two engaging elements **31** are arranged oppositely, each corresponding to a respective one of the two matching portions **11**. The elastic element **32** is mounted between the two engaging elements **31** and is capable of pushing the two engaging elements **31**. When the at least one assembling module **30** and the second component **20** are assembled to the first component **10**, the at least one assembling module **30** is disposed between the two matching portions **11**, so each of the two engaging elements **31** is pushed by one end of the elastic element **32** to abut against the corresponding matching portion **11**. The actuator **33** has two said first guiding inclined surfaces **331**, each corresponding to the second guiding inclined surface **311** of a respective one of the two engaging elements **31**; hence, operating the actuator **33** will push the two engaging elements **31** simultaneously. Alternatively, in another embodiment of the assembly structure in accordance with the present invention, the at least one assembling module **30** comprises only one said engaging element **31**. One of two sides of the second component **20** along the abutting direction D2 is assembled to the first component **10** through the engaging element **31**, and the other side of the second component **20** is assembled to the matching portion **11** of the first component **10** by tenoning. In either embodiment, the second embodiment can be easily attached to and detached from the first component **10**.

(24) Additionally, in a configuration of the assembly structure, the assembly structure comprises two said assembling modules **30**. The two assembling modules **30** are disposed at two opposite sides of the second component **20**, and the actuator **33** of each of the two assembling modules **30** can be operated to move toward a center of the second component **20**. That is, the force-applying directions D3 of the two assembling modules **30** are opposite to each other. The at least one engaging element **31** of each of the two assembling modules **30** corresponds to the at least one matching portion **11** of the first component **10** one to one. Alternatively, the engaging elements **31** of the two assembling modules **30** are disposed at one side of the second component **20** along the abutting direction D2, and therefore the engaging elements **31** of the two assembling modules **30** together cooperate with the at least one matching portion **11** of one side of the first component **10**. Alternatively, the engaging elements **31** of the two assembling modules **30** connect together in one piece, are abutted by the elastic elements **32** of the two assembling modules **30**, and are pushed by the actuators **33** of the two assembling modules **30**, hence increasing stabilities of operating the at least one engaging element and stabilities of connecting the second component **20** and the first component **10**. Alternatively, in another configuration of the assembly structure, the assembly structure comprises only one said assembling module **30**. The assembling module **30** is disposed at one of two ends, opposite to each other along the force-applying direction D3, of the second component **20**. The other one of the two ends of the second component **20** is assembled to the first component **10** by tenoning, or has no extra fixing structure so the second component **20** is assembled to the first component **10** only through the assembling module **30**, thereby decreasing costs.

(25) Specifically, the assembly structure is used in fields of lighting devices described as follows. With reference to FIGS. 4 and 11, an embodiment of a lighting device in accordance with the present invention comprises said assembly structure, at least one light guide plate **40** and at least one light-emitting unit **50**.

(26) With reference to FIG. 4, the at least one light guide plate **40** is mounted at the first component **10**. The at least one light-emitting unit **50** is mounted at the second component **20** or at the at least one engaging element **31** of the at least one assembling module **30** and faces toward the at least one light guide plate **40**. Besides, the at least one light-emitting unit **50** can be a light bar with multiple light emitting diodes (LED), or it can be a single light emitting diode depending on product requirements.

(27) In addition, the lighting device comprises a driver **60**, which is mounted at the second component **20**. The driver **60** is electrically connected to the at least one light-emitting unit **50** to power the at least one light-emitting unit **50**. Preferably, the driver **60** is mounted at a top side of the second component **20** to facilitate to connect to an external power source.

(28) With reference to FIGS. **11** and **12**, the first component **10** is a frame of the lighting device or is formed on or mounted at the frame of the lighting device. The driver **60** of the lighting device is mounted on the second component **20**. The second component **20** and the at least one assembling module **30** can be easily attached to or detached from the first component **10**, and the at least one light-emitting unit **50** and the driver **60** can be quickly attached to or detached from the frame of the lighting device. As the first component **10** and the at least one light guide plate **40** stay at their original positions without detaching, the convenience of assembling is increased effectively and the time of assembling is decreased. Transporting a whole set of the lighting device is unnecessary, therefore enhancing the transporting convenience. Also, since the at least one light-emitting unit **50** and the driver **60** can be detached and replaced individually, the first component **10** and the at least one light guide plate **40**, which are less likely to be damaged, can continue to be used. The second component **20** and the at least one assembling module **30** can also be reused. Hence the service life of the lighting device is prolonged and the waste is reduced, increasing recycling rates.

(29) In an embodiment of the lighting device, the first component **10** comprises an exterior frame **12** and at least one back plate **13**. The at least one back plate **13** and the at least one light guide plate **40** are mounted in the exterior frame **12**. The at least one back plate **13** is disposed on a top side of the at least one light guide plate **40**. The at least one matching portion **11** is formed at the at least one back plate **13**. When the second component **20**, the at least one assembling module **30**, the at least one light-emitting unit **50** and the driver **60** are detached from the first component **10**, the exterior frame **12**, the at least one back plate **13** and the at least one light guide plate **40** remain in their original assembling positions without being detached. The detachable components have much smaller volumes compared to non-detachable components, and therefore, on-site maintenance staffs can carry several sets of the detachable components to replace broken lighting devices, thereby increasing convenience.

(30) Additionally, because the second component **20** is assembled to the at least one back plate **13** mainly by the at least one engaging element **31** of the at least one assembling module **30**, and because the at least one light-emitting unit **50** is not mounted at the exterior frame **12**, the exterior frame **12** achieves a super narrow bezel design, thereby enlarging a light-emitting area of the lighting device. Furthermore, when the second component **20** is connected to the first component **10**, the fixing portion **336** of the actuator **33** extends into the exterior frame **12** of the first component **10** along the force-applying direction **D3**. The at least one engaging element **31** abuts against a top side of the at least one back plate **13** of the first component **10**, and the second component **20** abuts against a bottom side of the at least one back plate **13**. So, the second component **20** is simultaneously engaged and fixed by the fixing portion **336** of the actuator **33** and the at least one engaging element **31**. When detaching the detachable components, pushing the actuator **33** along the force-applying direction **D3** allows for detaching the fixing portion **336** of the actuator **33** from the exterior frame **12**. At the same time, the at least one engaging element **31** is retrieved inwardly to be moved away from the at least one matching portion **11** on the at least one back plate **13**. Therefore, the second component **20** is detached from a bottom side of the first component **10** along the assembling direction **D1**, so the second component **20** and the driver **60**, which is on the second component **20**, can be quickly detached.

(31) With reference to FIG. **4**, the at least one engaging element **31** has an installing chunk **315**. The installing chunk **315** extends along the force-applying direction **D3**, is parallel to a lateral edge of the at least one light guide plate **40**, and is adjacent to the lateral edge of the at least one light guide plate **40**. The at least one light-emitting unit is preferably mounted at the installing chunk **315** of the at least one engaging element **31**. Since the at least one engaging element **31** is pushed by the

elastic element **32**, the installing chunk **315** of the at least one engaging element **31** is pushed toward the lateral edge of the at least one light guide plate **40** along the abutting direction **D2**. Thus, the at least one light-emitting unit **50** mounted at the installing chunk **315** is directly contacted with the lateral edge of the corresponding light guide plate **40**. Therefore, the at least one light-emitting unit **50** directly emits light into the corresponding light guide plate **40**, effectively increasing the light-coupling effect and the light utilization rate.

(32) Moreover, if the at least one light guide plate **40** expands or shrinks due to temperature, by the elastic element **32** pushing outwardly along the abutting direction **D2**, a gap between the at least one light-emitting unit **50** and the at least one light guide plate **40** automatically reduces. Therefore, the at least one light-emitting unit **50** is still directly contacted with the corresponding light guide plate **40**, so the at least one light-emitting unit **50** and its corresponding light guide plate **40** will not shift away from each other. In turn, an angle of light-coupling remains unchanged, thereby keeping the light-coupling effect of the at least one light-emitting unit **50** and its corresponding light guide plate **40**. So, the light utilization rate of the lighting device will not be affected by the expansion or shrinkage of the at least one light guide plate **40** and thus has a higher brightness.

(33) Furthermore, in a configuration of the lighting device comprising only one said light guide plate **40**, the at least one assembling module **30** comprises only one said engaging element **31** and uses the engaging element **31** to push the at least one light-emitting unit **50** to be pressed tightly against its corresponding light guide plate **40**. In a configuration of the lighting device comprising two said light guide plates **40**, the at least one assembling module **30** comprises two said engaging elements **31**. The at least one light-emitting unit **50** is mounted at each of the two engaging elements **31**, which push the at least one light-emitting unit **50** toward its corresponding light guide plate **40**. Symmetrical design of the two engaging elements **31** achieves better connecting stability.

(34) Therefore, responding to market trends, the lighting device is applicable to the Lighting as a Service (LaaS) business model. For example, the operation and power-supply of the lighting device is monitored through the Internet of things any time and can be promptly repaired once malfunctioning is inspected. The replaced light device is directly recycled and reused by the manufacturer. On the contrary, replacing and discarding a whole set of the broken lighting device will produce a lot of wastes. Therefore, by the lighting device in accordance with the present invention, the manufacturers can move the prone-to-broken driver **60** from inside of the lighting device (i.e. the first component **10**), to components (i.e. the second component **20** and the at least one assembling module **30**), which are easily detachable from the lighting device. Therefore, the driver **60** can be replaced solely without discarding the whole set of the lighting device when broken, thereby prolonging the service life of the lighting device and decreasing the number of abandoned lighting devices that need to be disposed of.

(35) To sum up, the at least one assembling module **30** of the assembly structure makes the second component **20** easily detached from or attached to the first component **10**. When the at least one assembling module **30** is applied in the fields of lighting devices, the second component **20** and the driver **60**, which is mounted at the second component **20**, are easy to be detached, while the first component **10** and the at least one light guide plate **40** are not to be detached. Therefore, the assembly structure not only enhances the convenience of maintaining the lighting device, but also facilitates to transport and reduces wastes.

(36) Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. A lighting device comprising: a first component having at least one matching portion; a second component; at least one assembling module mounted at the second component and facilitating the second component to be detachably assembled to the first component along an assembling direction, and comprising two engaging elements movably mounted at the second component along an abutting direction, and wherein the two engaging elements are disposed at two sides opposite to each other of the second component along the abutting direction; and an elastic element mounted in the second component and disposed between the two engaging elements, and capable of pushing the two engaging elements along the abutting direction, such that one of the two engaging elements abuts against the at least one matching portion of the first component along the abutting direction to assemble the second component to the first component; wherein the assembling direction and the abutting direction are nonparallel; at least one light guide plate mounted at the first component; and at least one light-emitting unit mounted at one of the second component and the two engaging elements, and faced toward the at least one light guide plate.
2. The lighting device as claimed in claim 1, wherein the at least one assembling module comprises an actuator mounted at the second component and being movable along a force-applying direction, wherein when the actuator is operated to move along the force-applying direction, the actuator pushes one of the two engaging elements to depart from the at least one matching portion along the abutting direction, such that the second component is detachable from the first component along a direction being opposite to the assembling direction.
3. The lighting device as claimed in claim 2, wherein the at least one assembling module comprises a position-limiting plate mounted on the second component and covering the second component; the two engaging elements and the elastic element are disposed at an inner side of the position-limiting plate; the position-limiting plate has a position-limiting hole elongated along the force-applying direction; and the actuator has a position-limiting hook extending out from the position-limiting hole and abutting against an outer side of the position-limiting plate.
4. The lighting device as claimed in claim 2, wherein the actuator has at least one first guiding inclined surface; each of the two engaging elements has a second guiding inclined surface abutting against the at least one first guiding inclined surface; wherein when the actuator is operated to move along the force-applying direction, the at least one first guiding inclined surface of the actuator pushes the second guiding inclined surface of one of the two engaging elements.
5. The lighting device as claimed in claim 4, wherein the actuator has a handle extending out from a bottom side of the second component; and a guiding portion disposed in the second component; and the at least one first guiding inclined surface is formed at the guiding portion.
6. The lighting device as claimed in claim 5, wherein the second component has an opening hole formed at the bottom side of the second component; and at least one positioning protrusion formed at a periphery of the opening hole; the handle extends out from the opening hole of the second component; the actuator has at least one positioning recess adjacent to the handle; and the at least one positioning protrusion extends into the at least one positioning recess to limit the actuator in position along the assembling direction.
7. The lighting device as claimed in claim 4, wherein one of the two engaging elements has an engaging chunk elongated along the force-applying direction and abutting against the at least one matching portion of the first component; a guiding chunk mounted at the engaging chunk; and the second guiding inclined surface is formed at the guiding chunk.
8. The lighting device as claimed in claim 7, wherein the engaging chunk has a chunk guiding surface elongated along the force-applying direction, inclined with respect to the assembling direction, and abutting against the at least one matching portion of the first component.
9. The lighting device as claimed in claim 2, wherein the actuator has a fixing portion extending

along the force-applying direction toward the first component, such that the second component is limited in position by one of the two engaging elements abutting against the at least one matching portion of the first component along the abutting direction and is limited in position by the fixing portion of the actuator extending into the first component along the force-applying direction.

10. The lighting device as claimed in claim 1, wherein the lighting device comprises: a driver mounted at the second component.

11. The lighting device as claimed in claim 1, wherein the first component comprises: an exterior frame; and at least one back plate disposed on a top side of the at least one light guide plate; the at least one back plate and the at least one light guide plate are mounted in the exterior frame; and the at least one matching portion is formed at the at least one back plate.

12. The lighting device as claimed in claim 1, wherein each of the two engaging elements has an installing chunk extending substantially parallel to a lateral edge of the at least one light guide plate and being adjacent to the lateral edge of the at least one light guide plate; and the at least one light-emitting unit is mounted at the installing chunk of one of the two engaging elements.
